GPLOT V0.90 Command Cheat Sheet

SYSTEM

DEVICE	NAME [FILE [DIMENSIONS]] – Select output device [W,H[+X+Y]]
CLEAR	[FILE] – Clear drawing / end output file, change output file name.
FILL	Fill screen with colour (device dependent)
OBEY	NAME [PARAMETER–LIST] – Read commands from file (nestable)
HELP	TOPIC – Display help on topic
STATUS	Display status information
MEMTEST	Test dynamic memory. Generate graph plotting test data
GET	NAME – Get an indirect permanent file (ignored if not on NOS)
LOGFILE	NAME – Open a command log file
MAXPOINTS	NUMBER – Set max data points in internal arrays (default 1000)
NSTACK	N – Set evaluator stack size (minimum 4, default 8)
VERSION	[N] – Version information. Print or put in string register N
WAIT	Wait for user response (enter) on interactive device (eps/svg ignored)
RESET	Reset most state to defaults. Not device or read data.
PREFIX	PATH – Set a prefix to prepend to all filenames. ignored on NOS.
EXIT	Exit GPLOT

DEVICES

GTERM	Python/Qt/OpenGL based colour graphics terminal
TEK4K	Tektronix 4010/4014 terminal
EPSCOL	NAME [W,H+X+Y] – Encapsulated PostScript file (W,H,X,Y INCH)
	... default size 5,5+0.5+0.5 inches
SVG	NAME [W,H] – Scalable Vector Graphics file (W,H PIXEL)
	... default size 800,800 pixels

COLOURS AND STYLES

CSGROUP	ALL/GENERAL/TEXT/ANNOT – Colour/style group to set
COLOUR	R G B – Set rgb colour to use, range 0 to 1
WIDTH	WIDTH – Set line width
STYLE	STYLE SOLID/DASH/DOT/DASHDOT [LEN] – Line style, dash len (all groups)

GRAPH PLOTTING

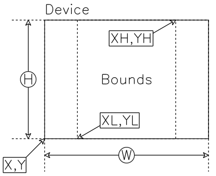
READ	NAME XCOL YCOL [YECOL [XECOL]] – Read a data file using columns.
	... NAME=HERE, use command input to EOF line
	... XCOL YCOL etc. are column numbers (1 ...) space or comma separated
XYPOINT	Draw xy graph with points
XYLINE	Draw xy graph with lines
XYHISTOGRAM	Draw xy histogram
GRMOVE	X Y – Move to graph coords (X,Y)
GRDRAW	X Y – Draw to graph coords (X,Y)
XYAUTO	Find both axis ranges automatically
XRANGE	XLO YHI – Set X axis range
YRANGE	YLO YHI – Set Y axis range
XYSAME	Keep previous X,Y axis ranges
XLINEAR	Use linear X axis
YLINEAR	Use linear Y axis
XLOG	Use log X axis
YLOG	Use log Y axis
INTVALUES	NONE/X/Y/BOTH – Try to use integers for axis values, if possible.
GRID	NONE/X/Y/BOTH – Grid graph in axis
INTERPOLATE	LINEAR/CUBIC/QUINTIC N – Interpolate with N intermediate points
ASYMERRORS	ON/OFF – Asymmetric Y error bars (off for X & Y error bars)
HISTSTYLE	ABUT/ABUT+SHADE/LINES/WIDE WIDE+SHADE [WIDTH] – Histogram style
ANNOTATE	ON/OFF – Turn graph annotation on/off
RIGHTANNOT	ON/OFF – Turn right edge annotation on/off (drawn only if annotate off)
TITLE	TEXT [LOWER] – Set title. If lower is yes, below instead of above
XLABEL	TEXT – Set X axis label
YLABEL	TEXT – Set Y axis label
RYLABEL	TEXT – Set right edge Y axis label
GSTYLE	BOXED/AXES/OPEN – Overall graph drawing style.
	... (axes must be in range to draw, see AXCUT)
AXCUT	X0 Y0 – Set point through which axes pass (default: 0,0)
MARKER	NUMBER – Set point marker number
USEKEY	Prepare to create a key (legend) for the graph
ADDKEY	TEXT – Add a key for the current graph line/points
KEYS	Draw the accumulated keys (after all lines/points)
GRAPHMODE	ON/OFF – Internally set bounds to same aspect ratio as the device.
	... if not ON, the graph appears in a centered, square, subregion
SUBFIGGRID	NX NY IX IY [SHRINK] – Set up pane to draw next graph as a sub-figure.
	... Figures arranged in NX X NY grid. this figure at IX,IY.

L–SYSTEM

LSYSTEM	NRULES NITER ANGLE – Generate and draw an L–system.
	... Axiom in STRING 1, rules in STRINGS 2–9
	... Alphabet: F A B C D E M X Y + – [ ]
F and D	Draw a unit length line segment at the current drawing angle.
M	Move by unit length at the current drawing angle.
+	Turn counter–clockwise by the turning angle.
–	Turn clockwise by the turning angle.
[	Push the current position and angle on to a stack.
]	Pop a position and angle off the stack and make them current.

BASIC DRAWING

BOUNDS	XL XH YL YH – Set plot bounds (user world coordinate system)
PANE	XL XH YL YH – Set pane (clipping) area.
	... also area in which graph will be plotted.
CANVAS	XL XH YL YH – Set bounds and pane (to be the same)
UNPANE	Stop using any pane
BLANK	XL XH YL YH – Set blank area
UNBLANK	Stop using any blank area
OUTLINE	PANE/BLANK/BOUNDS/DEVICE – Outline an area
MOVE	X Y – Move to position
DRAW	X Y – Draw to position
PATH	C/O – Draw a closed / open polyline.
	... coordinates from X,Y arrays (see over).
CURVE	C/O – Draw a closed / open smooth curve.
BEZIER	C/O – Draw a closed / open bezier curve.
CIRCLE	X Y R – Draw circle, center X Y, radius R.
ARC	X Y R A1 A2 – Draw a circular arc, center X Y, radius R
	... start angle A1, end angle A2.
RECT	X Y W H – Draw a rectangle, bottom left at X Y, wd W, ht H.
CRECT	X Y W H – Draw a rectangle, center X Y, width W, height H.
FONT	SET FONTNAME – Font/symbols/markers to use.
	... SET: 1 2 3 S M for 3 alphabets, symbol, marker.
LISTFONT	List available fonts
SYMHT	HEIGHT – Text, symbol or marker height in user bounds units
SYMANG	ANGLE – Text drawing angle wrt X axis (ccw degrees)
TEXT	TEXT – Draw text (with format control)
CTEXT	WIDTH TEXT – Text horizontally centered on current XPOS,YPOS,
	... scaled to WIDTH user units (0 for no scaling, use SYMHT)
MARKER	NUMBER YES – Draw marker at current position



HIGHER DRAWING – TEXT BOXES

BOXTEXT	TEXT [X Y] – Draw text in a box with centre or bottom left
	... AT X,Y or at the last incremented box position. Box
	... Position increments by specified deltas after each draw.
	... Text is centered vertically and centered or flush left
	... HORIZONTALLY. Text may have up to 5 lines separated by
	... BACKSLASHES.
BOXPSIZE	WIDTH HEIGHT BOTLEFT – Set box size. BOTLEFT YES/NO sets.
	... MEANING OF BOXTEXT X,Y NO = centre, YES = bottom left.
BOXPHATCH	SPACING IANGLE MODE – Hatch spacing, angle (in spacings), mode:
	... NONE/VERT/HORIZ/BOTH
BOXPTEXT	WIDTHFRAC MODE FLUSHLEFT – Set how text is drawn. MODE: FIX/SCALE.
	... WIDTHFRAC: Fraction of box width for text in SCALE mode.
BOXPBOX	NONE/OUTER/INNER/BOTH – Draw the box outline, inner box around text.
BOXPDELTAS	DX DY – Automatic box position step after every box draw.

HIGHER DRAWING – DECORATED LINES & LABELS

LINE	LINESPEC – Draw a decorated polyline defined by LINESPEC
	... LINESPEC is a sequence of POINTSPECs separated by) characters.
	... POINTSPEC is (type)X,Y[,A] where x and y are bounds coords.
	... (type) is P for a plain, undecorated, location, A for an
	... arrow with automatically determined direction, or S for a skip
	... (a shape to skip over an existing line). The optional A
	... is (very short) arbitrary text to place in a label in the middle
	... of the line segment the POINTSPEC starts.
	... Example: L I N E A 3 5 , 1 6) P 3 5 , 2 1 , R T) A 2 6 , 2 1
ARROWPARM	TYPE SIZE SHARPNESS BARBEDNESS – Arrow parameters. TYPE:
	... OPEN/CLOSED/BARBED, SIZE in bounds units, SHARPNESS 0.5:5,
	... BARBEDNESS fraction of arrow head in barb.
LINEPARM	SKIPSCALE ANNOTSCALE – Scale of skips and mid–line annotations.
GLABEL	GX GY LENGTH ANGLE TEXT – Draw a label with an arrow and boxed text.
	... GX,GY are graph range coordinates to which the arrow points,
	... LENGTH is the length of the arrow line in bounds units, ANGLE
	... is the angle of the line wrt +X in degrees, TEXT is the
	... text to appear in the label box.
ALABEL	X Y LENGTH ANGLE TEXT – As GLABEL but X,Y in bounds coords.

EVALUATOR (see over for operators)

ERANGE	BASE START STOP NELEM – Set range of values in X/0
	... stack register. BASE 1 linear, else log base.
EVAL	RPN [ARG] – Execute a procedure/evaluate a function using an RPN
	... NOTATION, ARG 1,2, etc. refer to proc registers using @1
ITEVAL	START END STEP RPN – Iterated evaluation over start to end by step.
2DEVAL	RPN – As EVAL, but preserve both X and Y data (see over).
2DITEVAL	START END STEP RPN – As ITEVAL, but preserve both X and Y.
PROC	N RPN – Store RPN code string in proc register N. Use EVAL @N
LOADPROC	N NAME – Load a named procedure from GPLPROC file into proc reg N
STRING	N TEXT – Set contents of string register N
STO	N X – Store real number X in memory register N
RCL	N – Display contents of memory register N
ZEROVAL	V – Set value for divide–by–zero traps. Default 1e–9. 0=no trap.
BBSTART	Accumulate bounding box instead of drawing with M & D operators.
BBEND	Exit bounding box mode and return to drawing with M & D operators.
BBSET	[SCALE] – Set bounds to match bounding box found with BBSTART/BBEND.

TEXT STRING MARKUP CODES

*U	Go to upper case
*L ... $L	Start/end lower case
*B	Backspace
*1 *2 *3	Select a font set
*+ ... $+	Start/end superscript (2 levels)
*– ... $–	Start/end subscript (2 levels)
*N	Reset sub/superscript level to 0
*O ... $0	Sub/super fully below/above last ch
*, NUM $, DEN $.	Construct a fraction
*:nn	Output symbol with code nn
*::nn	Output marker with code nn
*Vnn	Output alphabetic with code nn
*= ... $=	Start/end underlining

COMMAND LINE OPTIONS

GPLOT,KEY=VAL, ...	To run on NOS
UGPLOT KEY=VAL	To run on Unix–like systems.
OBEY=file	Name of obey file to run
PARM=string	Parameters for obey file in dbl qts
GET=yes or NO	Auto GET files on NOS
SAVE=yes or NO	Auto SAVE files on NOS
DEBUG=yes or NO	Show expanded parameters
QUIET=yes or NO	Show expanded command
SLIDE=YES or no	Adjust graph layout for larger text

PLNOTES MARKDOWN INTEGRATION

) GPLOT COMMAND
) GPLOT COMMAND
) ... etc.
) ... no DEVICE or CLEAR

(blank line ends GPLOT commands)

Use:
MODEXEC,BUILD,PLNOTES,ARG=ANOTE.
... to create ANOTE.HTML

GPLOT V0.90 Evaluator Cheat Sheet

OPERANDS

(DIGITS)	(-- C1)	SET TOS ARRAY TO A LITERAL CONSTANT.
X	(-- A1)	SET TOS TO X RANGE ARRAY.
PI	(-- C1)	SET TOS TO PI.
E	(-- C1)	SET TOS TO E.
I	(-- C1)	SET TOS TO ITERATION NUMBER FROM ITEVAL. 1 IF IN EVAL.
IDX	(-- A1)	SET TOS TO THE ARRAY ELEMENT INDEX.
TWPI	(-- C1)	SET TOS TO 2 PI.
PI/2	(-- C1)	SET TOS TO PI / 2

STACK MANIPULATION

SWAP or S	(A1 A2 -- A2 A1)	SWAP OR EXCHANGE TOP 2 STACK ARRAYS.
DUP or &	(A1 -- A1 A1)	DUPLICATE TOP OF STACK
G	(.. AN .. C1 -- AN)	GET STACK LEVEL AT DEPTH C1 INTO TOP OF STACK
POP	(A1 --)	POP TOP OF STACK
CL	(.. --)	CLEAR STACK
SETX	(A1 -- A1)	OVERWRITE RANGE, GRAPH X VALUES WITH STACK TOP VALUES: X = A1
SETY	(A1 -- A1)	OVERWRITE GRAPH Y VALUES WITH STACK TOP VALUES: Y = A1
XLIN	(C1 C2 C3 --)	X = C1 TO C2 IN C3 LINEAR STEPS
XLOG	(C1 C2 C3 C4 --)	X = C1**C2 TO C1**C3 IN C4 STEPS
EL	(A1 C2 -- A1 C2)	C2 = A1(C2)
I0IJ	(C1 C2 -- C1* C2*)	C1*=MOD(C1,C2), C2*=(C1/C2): 1D TO 2D INDEX CONVERT, 0 BASED.
I1IJ	(C1 C2 -- C1* C2*)	C1*=MOD(C1,C2)+1, C2*=(C1/C2)+1: 1D TO 2D INDEX CONV, 1 BASED.

BASIC ARITHMETIC

+	(A1 A2 -- A1)	ADD ARRAY ELEMENTS: A1 = A1 + A2
-	(A1 A2 -- A1)	SUBTRACT ARRAY ELEMENTS: A1 = A1 - A2
R-	(A1 A2 -- A1)	REVERSE SUBTRACT: A1 = A2 - A1
*	(A1 A2 -- A1)	MULTIPLY ARRAY ELEMENTS: A1 = A1 * A2
**	(A1 A2 -- A1)	EXPONENTIATION: A1 = A1 ** A2
/	(A1 A2 -- A1)	DIVIDE ARRAY ELEMENTS: A1 = A1 / A2
R/	(A1 A2 -- A1)	REVERSE DIVIDE: A1 = A2 / A1
RCP	(A1 -- A1)	RECIPROCAL A1 = 1.0 / A1
CHS	(A1 -- A1)	A1 = -A1
ABS	(A1 -- A1)	A1 = ABS(A1)

MATH FUNCTIONS

SIN	(A1 -- A1)	A1 = SIN(A1)
COS	(A1 -- A1)	A1 = COS(A1)
TAN	(A1 -- A1)	A1 = TAN(A1)
ASIN	(A1 -- A1)	A1 = ARCSIN(A1)
ACOS	(A1 -- A1)	A1 = ARCCOS(A1)
ATAN	(A1 -- A1)	A1 = ARCTAN(A1)
SINH	(A1 -- A1)	A1 = SINH(A1), DOMAIN |X| (742.36
COSH	(A1 -- A1)	A1 = COSH(A1), DOMAIN |X| (742.36
TANH	(A1 -- A1)	A1 = TANH(A1), DOMAIN |X| (742.36
SQRT	(A1 -- A1)	A1 = SQRT(A1)
LOG	(A1 -- A1)	BASE E LOGARITHM: A1 = LN(A1)
LG10	(A1 -- A1)	BASE 10 LOGARITHM: A1 = LOG10(A1)
LOG2	(A1 -- A1)	BASE 2 LOGARITHM: A1 = LOG2(A1)
EXP	(A1 -- A1)	A1 = E ** A1 DOMAIN: -675.81 TO 741.66
RAND	(A1 -- A1)	UNIFORMLY DISTRIBUTED RANDOMS [0:A1): A1 = A1 RANDOM
SEED	(C1 --)	SET RANDOM NUMBER SEED TO A1(1)
GCD	(C1 C2 -- C1)	FIND GREATEST COMMON DIVISOR OF C1, C2

NUMBER RANGE RELATED

MIN	(A1 A2 -- A1)	A1 = MIN(A1,A2)
MAX	(A1 A2 -- A1)	A1 = MAX(A1,A2)
MOD	(A1 A2 -- A1)	A1 = MOD(A1,A2) OR A1 = A1 % A2
SIGN	(A1 -- A1)	A1 = (A1 (0) ? -1 : 1

CONDITIONALS

ODD	(A1 -- A1)	A1 = 1 WHERE INT(A1) IS ODD ELSE 0
GT	(A1 A2 -- A1 A2 A3)	A3 = (A1) A2) ? 1 : 0 ;
LT	(A1 A2 -- A1 A2 A3)	A3 = (A1 (A2) ? 1 : 0 ;
LE	(A1 A2 -- A1 A2 A3)	A3 = (A1 (= A2) ? 1 : 0 ;
GE	(A1 A2 -- A1 A2 A3)	A3 = (A1)= A2) ? 1 : 0 ;
EQ	(A1 A2 -- A1 A2 A3)	A3 = (A1 == A2) ? 1 : 0 ;
NE	(A1 A2 -- A1 A2 A3)	A3 = (A1 != A2) ? 1 : 0 ;
NOT	(A1 -- A1)	A1 = (A1 == 0) ? 1 : 0 ;
SEL	(A1 A2 A3 -- A1 A2 A3)	A3 = (A3 == 0) ? A1 : A2 ;

GRAPHICS

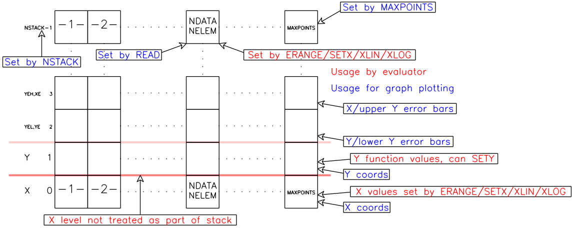
M	(C1 C2 --)	MOVE TO (C1,C2)
D	(C1 C2 --)	DRAW TO (C1,C2)
C	(C1 C2 C3 --)	DRAW CIRCLE, CENTER C1,C2 RADIUS C3.
A	(C1 C2 C3 C4 C5 --)	DRAW ARC, CENTER C1,C2 RADIUS C3, ANGLES C4 TO C5.
MK	(C1 --)	DRAW MARKER, CODE NUMBER C1, CENTER CURRENT POSITION.
BOX	(C1 C2 C3 C4 --)	DRAW RECTANGLE, BOTTOM LEFT C1,C2 WIDTH C3 HEIGHT C4.
HDSH	(A1 -- A1)	DRAW RELATIVE (1,0) IF A1[I]) 0 ELSE MOVE
VDSH	(A1 -- A1)	DRAW RELATIVE (0,1) IF A1[I]) 0 ELSE MOVE
PTHO	(A1 A2 --)	DRAW POLYLINE X=A1,Y=A2, OPEN
PTHC	(A1 A2 --)	DRAW POLYLINE X=A1,Y=A2, CLOSED
CRVO	(A1 A2 --)	DRAW SMOOTH CURVE X=A1,Y=A2, OPEN
CRVC	(A1 A2 --)	DRAW SMOOTH CURVE X=A1,Y=A2, CLOSED
BEZO	(A1 A2 --)	DRAW BEZIER CURVE, CONTROL POLYGON X=A1,Y=A2, OPEN
BEZC	(A1 A2 --)	DRAW BEZIER CURVE, CONTROL POLYGON X=A1,Y=A2, CLOSED
T	(C1 --)	DRAW CONTENTS OF STRING REGISTER C1 AS TEXT.
TVF	(C1 C2 -- C1)	DRAW C1 AS TEXT USING FORMAT STRING IN C2.
TVI	(C1 C2 -- C1)	DRAW INT(C1) AS TEXT USING FORMAT STRING IN C2.
TS	(C1 --)	DRAW SYMBOL WITH CODE NUMBER C1
TM	(C1 --)	DRAW MARKER WITH CODE NUMBER C1
TC	(C1 --)	DRAW CHARACTER WITH CODE C1 FROM CURRENT ALPHABET
TH	(C1 --)	SET TEXT/MARKER/SYMBOL HEIGHT
TA	(C1 --)	SET TEXT DRAWING ANGLE IN DEGREES CCW
TSC	(--)	START TEXT CONTINUATION
TEC	(--)	END TEXT CONTINUATION
TLEN	(C1 -- C1)	GET THE LENGTH IN BOUNDS AT UNIT HEIGHT OF STRING IN REGISTER C1.
ROT	(A1 A2 C3 -- A1 A2)	ROTATE X=A1,Y=A2 BY C3 RADIANS ABOUT ORIGIN.
TRN	(A1 A2 C3 C4 -- A1 A2)	TRANSLATE X=A1, Y=A2 BY C3,C4
SCL	(A1 A2 C3 C4 -- A1 A2)	SCALE X=A1, Y=A2 BY C3,C4
LAB	(C1 C2 C3 C4 C5 --)	LABEL AT (C1,C2), LEN C3, ANG C4, STRING REG C5

NUMBER TYPE CONVERSIONS

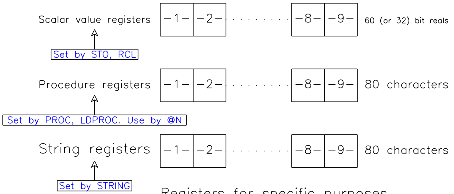
INT	(A1 -- A1)	TRUNCATION A1 = INT(A1)
FRAC	(A1 -- A1)	FRACTIONAL PART A1 = FRAC(A1)
FLR	(A1 -- A1)	A1 = FLOOR(A1)
CEIL	(A1 -- A1)	A1 = CEILING(A1)

SYSTEM

STO or =	(A1 C2 -- A1)	STORE A1(1) (A.K.A. C1) IN REGISTER C2
RCL or (HASH)	(C1 -- C1)	RECALL CONST VALUE FROM REGISTER INT(C1)
P	(--)	PRINT TOS ARRAY FIRST AND LAST ELEMENTS
PE	(A1 C2 C3 -- A1)	PRINT ELEMENTS C2 TO C3 OF A1 IN FREE FORMAT
PC	(C1 -- C1)	PRINT C1 (TOS A[1]) IN FREE FORMAT.
DUMP	(--)	PRINT ALL STACK LEVELS FIRST AND LAST ELEMENTS
R2D	(A1 -- A1)	CONVERT RADIANS TO DEGREES
D2R	(A1 -- A1)	CONVERT DEGREES TO RADIANS



Stack of arrays used both for graph point data and by evaluator



Registers for specific purposes